AI-POWERED ANALYSIS FOR INTELLIGENT HEALTHCARE ASSISTANCE

A retrieval-augmented medical chatbot with domain grounding and cloud deployment

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RAG | MEDICAL QA | AWS

TAKE-HOME MESSAGE

Retrieve the right evidence first, then let the language model explain it.

This prototype connects a 637-page reference and 5,859-vector medical index to GPT-4o through semantic search and a deployable Flask interface.

5,859	**500 / 20**	**384**	**TOP 3**	**<= 3**
indexed vectors	chunk size / overlap	embedding dimensions	retrieved passages	sentences per answer

THE RESEARCH QUESTION

Can a large medical reference be transformed into fast, focused answers without fine-tuning a model?

OBJECTIVE

Build an end-to-end RAG assistant that retrieves domain context, generates a concise answer and serves it through a web application.

DATA TO KNOWLEDGE

One source, prepared once, then reused for every question.

1. **SOURCE** The Gale Encyclopedia of Medicine, 2nd ed. - 637 PDF pages
2. **LOAD** PyPDFLoader extracts page text; metadata is reduced to source only
3. **SPLIT** Recursive chunks of 500 characters with 20-character overlap
4. **EMBED** all-MiniLM-L6-v2 maps each chunk to a 384-dimensional vector
5. **INDEX** Pinecone stores 5,859 dense vectors for cosine search in AWS us-east-1

RAG keeps the knowledge layer replaceable and avoids model fine-tuning.

HOW ONE ANSWER IS MADE

Two connected flows: an offline knowledge build and an online question-answering path.

BUILD ONCE | INDEX

PDF **MEDICAL REFERENCE** 637 pages → TXT **EXTRACT + SPLIT** 500 / 20 characters → EMB **MINILM ENCODER** 384-dimensional vectors → DB **PINECONE INDEX** 5,859 records

EVERY QUESTION | RETRIEVE + GENERATE

YOU **USER QUESTION** submitted to /get → Q **QUERY VECTOR** same MiniLM encoder → K=3 **TOP MATCHES** Pinecone retrieval → LLM **GPT-4o + CONTEXT** LangChain RAG prompt → A **CONCISE ANSWER** returned to web chat

PROMPT GUARDRAIL

Use retrieved context. If the evidence is insufficient, say "I don't know." Keep the answer to three sentences maximum.

Retrieval narrows a large corpus to the most relevant evidence; generation turns that evidence into a readable response.

SAFETY, LIMITS AND NEXT STEPS

EDUCATIONAL SUPPORT ONLY

Not a diagnosis, prescription or emergency service. Users should not enter identifying...

CURRENT LIMITATIONS

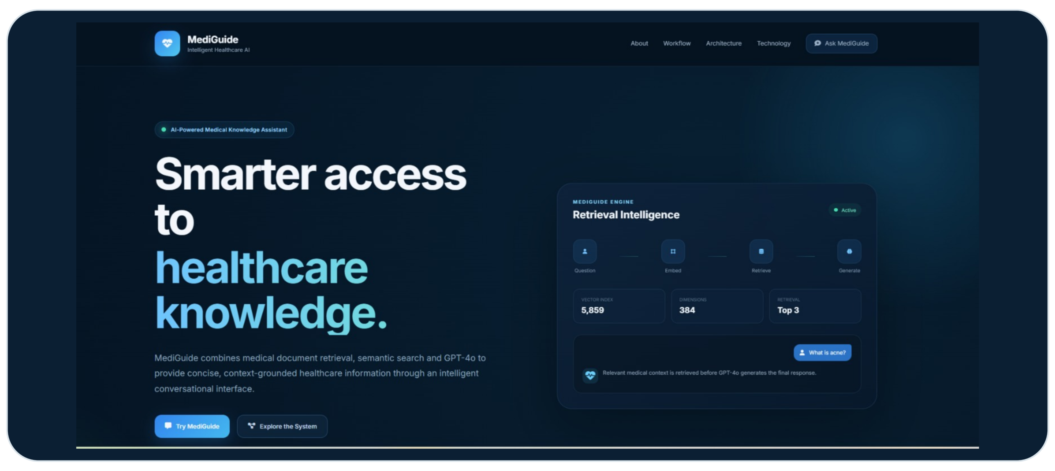
- The medical source is from 2002 and may not reflect current guidance.
- The 12-question benchmark tests retrieval only; it is not a clinical answer evaluation.
- No evidence citations or confidence display; current code also prints queries to logs.

NEXT EVALUATION

1. Use current clinical sources and return page citations.
2. Extend evaluation to clinician-reviewed questions, faithfulness, latency and safety.
3. Add red-flag escalation, privacy, authentication, rate limits and production logging.

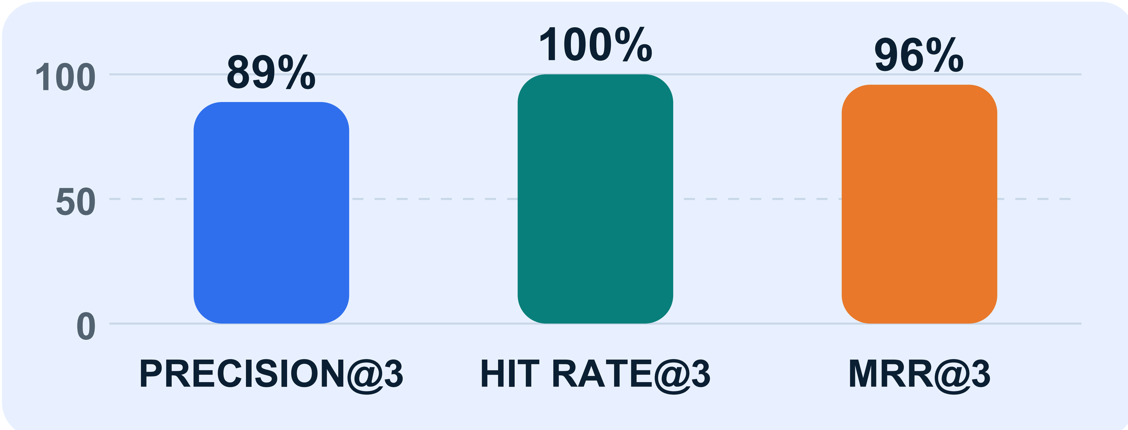
PROTOTYPE AND RETRIEVAL EVALUATION

MEDIGUIDE INTERFACE



Implemented web interface for exploring the indexed medical knowledge base.

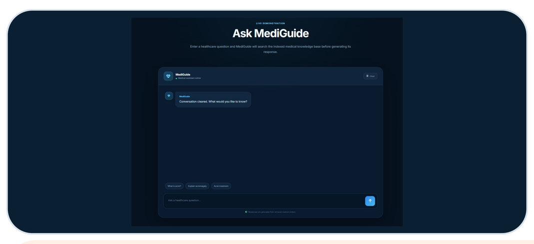
PRELIMINARY RETRIEVAL CHECK



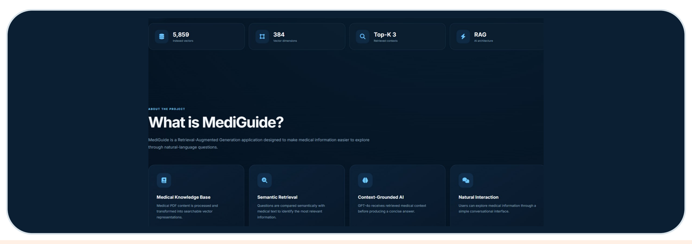
n=12 topic-focused prompts; binary topic-label relevance. Retrieval only - not clinical accuracy.

IMPLEMENTATION EVIDENCE AND DELIVERY

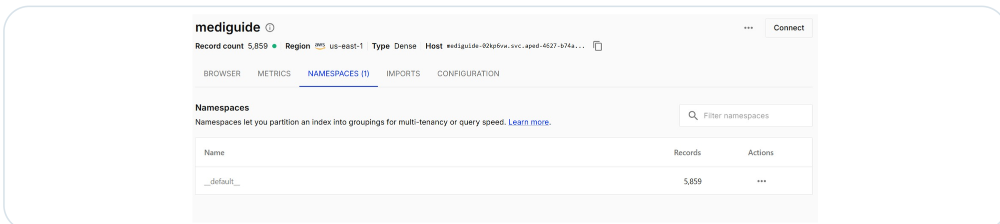
LIVE CHAT



SYSTEM SUMMARY



PINECONE: 5,859 RECORDS



DELIVERY: GitHub Actions -> Docker -> Amazon ECR -> Amazon EC2 -> Flask :8080

CONCLUSION

RAG provides a practical route from a large reference PDF to a usable medical information assistant. The prototype demonstrates the complete software pipeline; clinical use requires current evidence, rigorous evaluation and stronger safety controls.

REFERENCES

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